If you answered SOMETHING for each question, you had a good chance at passing. Practice exam technique – write a basic answer for each and return to flesh it out, rather than leaving stuff out completely and focusing a lot on some questions worth not much…

Section 1

Question 1

No solutions needed to be explained – ones on memo just for future understanding

A) Select either 1 AND 2 or 3 = 1 mark. Lose a mark for iv and half mark for just i and ii

The rest are just correct answer = 1 mark

Question 2

A)

- +1 mark: recursively partition (use some kind of **partitioning scheme)**
- +1 mark: partitioning makes it so that left half of array has one property and right half has opposite quality
- Just **mentioning** quickselect = 0.5 mark ONLY
- No marks for finding the smallest element – must be **kth-smallest**
- 1 mark for some kind of working algorithm that WOULD BE IMPLEMENTABLE for A
    - 0.5 if it's too vague
- ½ mark for max heap

Common mistakes:

- Finding the **absolute smallest** and not **k-th smallest** (0 marks)
- Mentioning binary search (the array is not sorted…)
- Having some weird algorithm which is poorly explained (only 0.5 marks)
- Algorithm which is equivalent to presort (0 marks unless it's creative enough, like maxheap which is technically a bit different, then 0.5 mark)
- Having a weird, slow algorithm but is well explained (1 mark)
- Using a max heap – this is equivalent to a presort (0.5 mark)
- Looping through array or similar… this is a bad algorithm

    - 0.5 marks if your algorithm is vague
    - 1 if it is specific enough to work
- For quickselect algorithms, not being specific about the property that the partitioning enforces (one half <p and one >p)
- For quickselect, not mentioning partitioning and only pivoting, or not saying that it is repeated or recursive
- Common mistakes:
- mix of old and new notation for lines (B/R vs B;B;R)
- Removing 2 1s from the end instead of 1.

B)

- 1 mark: Just note that it can degenerate to O(n^2) or Theta(n^2)
- 0.5 mark: Just note that it can degenerate (but without listing bound)
- Don't need to explain that sort is O(n log n) or anything...
- Stating that it's O(n^2) even w/o quickselect mentioned = 0.5 mark
- "If the kth smallest is the first or last element" for quickselect could be 0.5 mark (w/o mentioning time complexity)

Common mistakes:

- Explaining why your bad algorithm in A) is worse... we know it's bad... this is not new information. You could just design a purposefully bad algorithm to try get marks on B for explaining how it's bad
    - But if you're specific enough then you can get 0.5 mark
- The question is about **time complexity**, not **correctness**. If you say "my algorithm could have cases where it doesn't work", your answer for A is wrong and B is not answering the question...
- Not comparing to presort as is asked in the question (0 marks unless you mention that the worst case is O(n^2), then 0.5 marks)
- Not being specific about time bounds (0.5 marks)
- **SORTING IS N LOG N**
- The time complexity of the pre-sort has to be included in the final time complexity

Question 3

Pretty much just to test that they know about interpolation search

- 1 mark: Emphasise that the **interpolation cause uneven split** (don't even need to be clear how)
- 0.5 mark: Vague inkling that interpolation is causing **a** problem (but not specific)
- 1 mark: contrast with binary search which always chooses the middle. Must state to get the mark

Common mistakes:

- Must **compare** with the binary search, not just explain

Question 4

- The algorithm is basically just to split numbers in two halves and recursively sum
- 0.5 marks if they can vaguely show it's linear and not logn (but don't define recurrence)
- Most are missing a base case – need to mention
- If evidence of dividing array by 2, can award 1/2 for splitting array (from first part)
- Most just use the formulae and do not state the recurrence relationship, can award extra ½ mark for splitting as above if A(n/2) -> suffix not required
- The A(n) = 2 (n/2) + 1: First part is combining halves, while +1 includes the add adding the halves
- Common Mistakes:
    - Not defining problem and recurrence relationship before plugging values in the master theorem
    - Not defining a base case
    - Not including additional add operation combining problem halves
    - Not taking into account odd array length – why floor and ceil is needed (directly from textbook)

Question 5

- Don't need to say "not in O(log n)" explicitly
- ZERO marks if they just say O(n) or Theta(n) - could just be a guess. Must show **some** kind of understanding

Question 5

- **NEED** the binary number conversion, otherwise no mark for that – this is how they've been taught it… if they just do decrease and conquer then it isn't Horner's rule

- Don't technically need the table – just show that, by the binary representation, you get 5 multiplications
- 1 mark if they just state the answer as 5
- If the algorithm is correct/demonstrate understanding but answer is wrong then they can get 1 mark

Common mistakes

- Just showing it by repeated grouped multiplies – needs to be binary
- Showing code did not answer the question and did not get you marks.

Question 6

- The "trick" is not to start at the very last element of the table… start at **COLUMN j=5**. If they start at last element, cap at 2 mark
- If their final answer is wrong but they apply the algorithm correctly, they will get no mark for that part but you can give marks with error carrying – but capped at 2

Common mistakes:

- Not changing column when an item is added
- Changing column based on the value, not the weight – still got to the right answer in this case but not always correct. Capped at 2
- Just writing down an algorithm but not applying it to this question
- Using brute force logic, the wrong algorithm, or no algorithm to get to the answer

Section 2

Question 7

No half marks in this question at all

A)

- Solution should go past all the 1s, flip the separator B, go past rest of 1s and then change last 1 to a B
- +3 marks for being able to draw **any correct TM syntax –** see the memo for specific marks
- +1 mark for replacing B with 1
- +1 mark for replacing last 1 with B - (doesn't depend on replacing B with 1)
- If in doubt, put aside

- If you choose to use the Turing Machine notation taught last year (where there are 3 components on each arc), then you can _*only*_ use that notation, you cannot sometimes use that notation and sometimes use the notation taught this year.
- If 2 different notations are used for a Turing Machine exam answer, you will not get any marks. Choose one notation and use only that.

B) should be the number of arrows (incl. self-returning arrows, NOT including start state arrow in) – **mark their TM, not the memo**. So count number of arrows in their TM

Common mistakes:

- Including the start state arrow in in the count

C) Give any one arrow and give the four correct pieces of info. Start state, read symbol, write symbol, next state. In **any order. But NO HALF MARKS**

Question 8

- Memo answer's square brackets should be ALL ROUND BRACKETS
- If they use square brackets it's only really correct for [01] instead of (0|1) most likely, as [abcdefg] means "match ONE instance of any character in 'abcdefg')
- +1 mark for the (1|0) or (0|1)
- +1 mark for the +
- +1 mark for the = being optional (so, for example, <=? or (<=|<))

Common mistakes:

- Many students using the * (Kleene closure) symbol rather than + (positive closure)
- Students using square brackets rather than round
- Students not correctly using the OR pipe for (1|0) – using (1.0)
- Incorrect placements of kleene closures

Question 9

- +0.5 mark: Any TM can be represented by binary number
- +0.5 mark: that makes them countable / natural numbers are countable (etc)

Common mistakes:

- Students mentioning real numbers as being countable
- Students using the diagonalization theorem
- Students mentioned the infinite tape as a reason for there being countable infinite TM

Question 10

Now out of 3, NOT 4. **B doesn't exist!!!!! So test total is now 34**

A & D are false, NOT C. No arguments about it. So they would get 2.5 marks

Q10: none = 1 mark (but blank = no marks)